

Question 1: SLIC: appearance versus spatial proximity

Choose **ONE** answer

You run SLIC twice on the same image, keeping the requested number of superpixels fixed. In the second run, you substantially increase compactness. What change would you generally expect?

- A. More superpixels, each covering a smaller region.
- B. More regularly shaped superpixels, potentially following object boundaries less closely.
- C. Superpixels that follow colour boundaries more closely, regardless of their shape.
- D. Superpixels that correspond more closely to complete semantic objects.

Answer 1: SLIC: appearance versus spatial proximity

Answer and explanation

Correct answer: B

More regularly shaped superpixels, potentially following object boundaries less closely.

Higher compactness gives spatial proximity greater weight relative to colour similarity. SLIC therefore tends to produce more regular regions, which may cross an object boundary when the colour evidence is weak.

Why the other options are incorrect

- A: The requested number of superpixels is held fixed. The final count may vary with connectivity cleanup, but increasing it is not the role of compactness.
- C: This reverses the trade-off. Increasing compactness puts more emphasis on spatial proximity.
- D: SLIC groups pixels by appearance and position. It does not recognise semantic objects.

Compactness controls a trade-off between spatial regularity and following image boundaries.

Question 2: Why do skip connections help?

Choose **ONE** answer

An encoder–decoder combines high-resolution encoder features with decoder features through skip connections. You randomly shuffle the spatial positions of the encoder features before combining them, preserving their values and tensor dimensions. What is the most likely consequence?

- A. Nothing changes, because the same feature values are still available.
- B. The output segmentation becomes smaller because spatial information has been removed.
- C. Boundary localisation becomes less reliable because encoder and decoder features no longer correspond to the same locations.
- D. The network can no longer produce a class prediction for every pixel.

Answer 2: Why do skip connections help?

Answer and explanation

Correct answer: C

Boundary localisation becomes less reliable because encoder and decoder features no longer correspond to the same locations.

Skip connections supply spatially aligned detail. After shuffling, a decoder location may receive encoder features from an unrelated part of the image, making boundary localisation less reliable.

Why the other options are incorrect

A: Feature location matters in dense prediction. The same values in different positions do not provide the same information.

B: Tensor dimensions are unchanged. Shuffling does not itself change the output resolution.

D: The network can still produce a class prediction at every output location, although predictions may be less accurate.

Useful skip features must preserve where the image evidence came from.

Question 3: Can different mistakes receive the same score?

Choose **ONE** answer

A ground-truth object occupies 100 pixels.

Prediction A selects 50 pixels, all inside the object.

Prediction B selects 200 pixels, including all 100 object pixels.

Which statement is correct about their foreground IoU?

- A. Both score 0.50, although A misses foreground and B includes extra background.
- B. A scores 0.50 and B scores 1.00, because B finds the entire object.
- C. A scores 1.00 and B scores 0.50, because every pixel selected by A is correct.
- D. Both score 0.67, because IoU counts the overlap twice.

Answer 3: Can different mistakes receive the same score?

Answer and explanation

Correct answer: A

Both score 0.50, although A misses foreground and B includes extra background.

IoU measures intersection divided by union. Prediction A has intersection 50 and union 100, so $\text{IoU} = 0.50$. Prediction B has intersection 100 and union 200, so $\text{IoU} = 0.50$.

Why the other options are incorrect

B: Finding all foreground pixels gives perfect recall, but false positives still reduce IoU.

C: Selecting only correct pixels gives perfect precision, but missed foreground still reduces IoU.

D: Counting the overlap twice is part of Dice, not IoU. Both predictions have $\text{Dice} = 2/3$, approximately 0.67.

Equal IoU can hide different errors. Inspect the masks as well as reporting the score.